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Back-to-Basics Hospital-wide Training Improves Inpatient Pediatric Care Experiences

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Abstract

Objective. Hospital leaders strive to improve care experiences. Many use training sessions to improve clinician-patient interactions.

Methods. We examined whether a hospital-wide Back-to-Basics training for clinical and non-clinical providers that focused on communication behaviors and safety protocols improved inpatient pediatric care experiences. The curriculum utilized didactics, video demonstrations, and roleplay through interactive, fast-paced stations delivering content tailored to employee role. Child HCAHPS scores were compared 24 months before (n=544) and 13 months after the training (n=643) to assess changes in inpatient pediatric care experiences.

Results. We observed a statistically significant increase in the overall mean top-box scores after Back-to-Basics implementation for five (of eight targeted) Child HCAHPS measures: *Communication between you and child's nurses* (3 percentage point (pp) pre-post increase, p-value=0.04), *How well nurses communicate with child* (7pp increase, p-value=0.001), *How well doctors communicate with child* (6pp increase, p-value=0.007), *Preventing mistakes and helping report concerns* (4pp increase, p-value=0.03), and *Helping child feel comfortable* (4pp, p-value=0.03). We found no significant differences in models that tested whether the effect of Back-to-Basics differed by service line or by patient racial, ethnic and language groups.

Conclusions. A small investment of curricular time devoted to communication skills and safety procedures across all staffing roles over three months improved Child HCAHPS communication-specific and safety scores. Since this training could be implemented in a variety of pediatric and inpatient settings to improve patient and family care experiences, further evaluation of the

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Contributors Statements:

Dr. Quigley conceptualized and designed the study; gained funding acquisition; led partnership and acquisition of the data; analyzed and interpreted the data; drafted the article; revised the article critically for important intellectual content; and was involved in final approval.

Dr. Panayotou supported the development and conduct of the training; revised the article critically for important intellectual content; and was involved in final approval.

Dr. Slaughter led the data analysis, advised on the draft of the article, analyzed and interpreted the data, provided critical input and revisions, and was involved in the final approval.

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intervention in different pediatric hospital sites, settings (i.e., hospital-within-a hospital), and contexts (urban, rural) is warranted.

Keywords

pediatric inpatient care; patient experience; training; communication; safety

Introduction

Healthcare organizations want to improve patient care and experiences.[1, 2] All hospitals and many children's hospitals measure inpatient care experiences using the Consumer Assessment of Healthcare Providers and Systems (CAHPS®) patient experience survey data. Hospital leaders use patient experience data to provide insights on their quality of care and to identify areas for improvement across site locations and providers within an organization.[3–6]

Healthcare organizations generally use one or more methods of changing physician, nursing and clinical staff behaviors, including education and training, feedback, engaging clinicians as leaders in change efforts or administrative changes, and using financial incentives, or financial penalties.[7, 8] The feedback method is often used as it can provide specific insights for a given provider; this entails hospital leaders reviewing performance data at the clinician or team level, assembling insights, and communicating potential improvements to providers or teams as a method of training.[9–11] Another common method is to conduct training sessions over a period of time for clinicians and non-clinicians on specific topics; this has been shown to be a promising intervention for improving behaviors and skills of attending physicians[10, 11] (e.g., training and use of a 'surgical flight plan' to standardize communication to patients;[12] 8-hour experiential communication skill training for physicians;[13] training on etiquette-based behavior such as knocking on a patient's door, asking to enter the patient's room, introducing oneself, sitting down in patients' room, and explaining one's role in care)[14] and for residents and nurses (e.g., 2-hour didactic and roleplaying training sessions on bedside communication skills and recognizing low health literacy among inpatients; nurse training on PEEP rounds (pain, elimination, environment, position);[15] nurse training on pain assessment after surgery[16]).[17–19]

Substantial time and resources are invested in providing training for clinicians (such as continuing medical education). However, little research has been done to test the effect of such training on patient experience scores and those studies that have provide little evidence of its effectiveness.[20, 21] We examine whether a hospital-wide training for clinical and non-clinical staff tailored to role improves inpatient pediatric care experience scores and if improvements are consistent across patients and service lines.

Methods

Setting.

We partnered with an urban/suburban, 216-bed freestanding children's hospital with 1,155 non-clinical staff and 1,300 clinicians with annual admissions of 3,400 patients and over

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40,000 emergency room visits located in the South in the state of Florida. We obtained de-identified Child HCAHPS surveys completed in English and Spanish from parents or guardians of hospitalized children under the age of 18 with discharge dates from May 1, 2021 through October 15, 2024 (6.4% response rate; 4.3% undeliverable rate; both comparable to the American Hospital Association region 4 response rate of 6.3% and lower than the national response rate of 8.7%).

Intervention.

The pediatric hospital Quality department wanted to conduct a hospital-wide training because of the high staff turnover, many staff who were new to caregiving, and the identified need to focus on safety and improve areas of care identified through a review and analysis of the hospital's Child HCAHPS data. The hospital's nursing turnover rate in 2023 during the discussions leading up to the training was 15.5%, which had increased and was approaching the 2023 national nursing turnover rate of 18.8%.[22] The hospital nursing and patient experience leaders partnered with the study team to examine the hospital's annual performance of their Child HCAHPS data. Typically, the hospital leaders reviewed the overall unadjusted Child HCAHPS composite measure performance (high/low performance and against benchmarks) to determine areas needed for improvement but were interested in understanding the underlying patterns of performance and importance to patients and families (i.e., correlation to overall hospital rating) overall, and by their main patient subgroups. The examination of these two patient experience metrics, i.e., case-mix adjusted mean score for each patient experience measure and the partial correlation of that measure with the overall rating, described in detail elsewhere,[23] pointed to the need for improvement in the areas of communication with parent and nurses, communication with parent and doctors, being kept informed, and helping the child feel comfortable and preventing mistakes (i.e., 8 Child HCAHPS measures). Focused improvements were not needed for the 2 global ratings or for several Child HCAHPS domains (i.e., 8 Child HCAHPS measures): responsiveness to call button, preparing to leave the hospital, communication about medicines, pain, involving teens, privacy, and hospital environment.

Back-to-Basics was a hospital-wide training implemented with clinical and non-clinical providers September 20, 2023, through November 2023. Overall, 1,005 employees attended out of a total possible 1,283 employees (~78% attendance; denominator does not take into account employees out on leave, out sick, or on vacation and does not include contracted clinicians). The training utilized didactics, video demonstrations, and role play through interactive, fast-paced stations (role determined stations) delivering specific content that included desired behaviors related to keeping patients/families informed about care, tests, procedures, communication with parents (i.e., listening techniques), communication with children (explanations, techniques for encouraging child to ask questions), helping the child feel comfortable, handling challenging situations, caring for those with non-English language preferences, the use of medical interpreters and safety protocols (i.e., infection control/hand hygiene, preparation of surgical patients, scrubbing the hub, and pressure injuries). Nine sessions were offered twice a day for 3 hours. Each session was scripted and taught by subject matter experts employed at the hospital with defined learning outcomes. Every participant attended a one-time three-hour training session that included nine topics

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or modules tailored to their role. For example, all clinical, front-line providers rotated through all 9 modules, whereas non-clinical staff attended 4 (of the 9) modules (i.e., patient experience, environment of care, journey to excellence/magnet, and National Database of Nursing-Sensitive Quality indicators and patient experience data). Table 1 delineates the nine modules describing the content, delivery style, learning objectives, and the included roles (i.e. clinical, non-clinical, both).

Measures.

The Child HCAHPS survey includes 10 composite measures, 2 global rating measures, and 6 single-item patient experience measures. Details can be found at https://www.ahrq.gov/sites/default/files/wysiwyg/cahps/surveys-guidance/hospital/about/child-hcahps-english-survey-954a.pdf. For each of these measures, we calculated the mean case-mix adjusted top-box score. The top-box indicates the proportion of responses in the highest response option category (e.g. “Always” for Never to Always response scales), except for a 0-10 hospital rating, for which the top two responses (9 and 10) were considered the top-box.[4] The main study intervention of interest is the implementation of the Back-to-Basics program, which began September 20, 2023, so we evaluated changes from surveys completed on or after this date.

Data and Analysis.

The sample included 1,187 completed surveys with 544 completed surveys pre-intervention and 643 completed surveys post-intervention.

To calculate adjusted mean scores, we first constructed case-mix adjustment variables: age of child (administrative), self-reported respondent’s age (<25 yrs old, 25 to 34, 35 to 44, 45 to 54, 55 to 64, 65 or older), education (Less than High School, High School graduate, Some college, 4-year college graduate), relationship to child (Mother, Other than Mother), and child’s general health (Excellent, Very Good, Good, Fair/Poor). These simplified categorizations were necessary due to the small sample sizes in some categories. Missing data, ranging from 1.1% to 2.8% among case-mix variables, was imputed using the overall mean.

We also combined the self-reported preferred language of the respondent and the child’s race and ethnicity, here forth referred to as race, ethnicity and language, to create four groups: English-preferring Hispanic, Spanish-preferring Hispanic, non-Hispanic Black, and non-Hispanic White patients. If self-reported data was missing for race, ethnicity and language (N=62), we used the administrative information from patient electronic medical records that was linked by the survey vendor to each survey response for survey language and race/ethnicity if available. With guidance from the hospital’s clinical leadership, we reviewed the service line variable that is linked to the survey responses. This variable is an administrative variable that is automatically assigned to each survey sent out for completion based on the unit from which the patient is discharged from their pediatric stay. However, if a patient spent 6 days in the intensive care unit (ICU) and spent 2 days and was discharged from an acute care unit, the variable would assign the patient to the ICU service. We combined the 33 types of services provided to the child during their hospital stay into four

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service line categories: pediatric general, pediatric intensive/critical care, pediatric surgeries and pediatric specialties.

There were surveys from 81 respondents excluded from our analytic sample because they either had 1) missing data for service line, administrative or self-reported race/ethnicity or preferred language, or 2) the race, ethnicity and language group sample size was too small or heterogenous to make statistical comparisons (i.e., we excluded 42 multiracial, 3 Native American, 22 Asian, 3 Hawaiian or pacific islander, 3 unknown, 3 black with primary language Haitian-Creole). We fit the models with and without these respondents excluded and findings were similar (data not shown). As a result and for consistency, the analytic sample therefore consisted of 1,197 surveys.

For each of the Child HCAHPS measures, we examined linear models that included time since Back-to-Basics was implemented, modeled as days since September 20, 2023, with both linear and quadratic terms, to see if the effect of the Back-to-Basics program had a complex relationship over time. We evaluated this complex time model with a simpler (nested) model that only included a dichotomous variable indicating before and after Back-to-Basics implementation.

We examined models unadjusted and adjusted. Adjusted models included the full set of case-mix adjuster variables, service line group, and the race, ethnicity and language variable. Using an analysis of variance (ANOVA) approach, we compared the residual sum of squares from these two models, i.e. whether the complex time effect improved model fit. None of the models had a significant F-test from the ANOVA, i.e. all F-test p-values were > 0.05, and therefore we used the simpler model that used only the dichotomous variable indicating time before and after Back-to-Basics as our final models.

Final ordinary least squared (OLS) models were used to calculate mean case-mix adjusted top-box scores before and after Back-to-Basics implementation. From this model, we calculated a t-statistic by dividing the estimated effect of the Back-to-Basics period with its standard error. The null hypothesis was that the Back-to-Basics period effect was zero and p-values from the t-statistic less than 0.05 were considered statistically significant. We did not adjust for multiple testing as this analysis was exploratory. We also examined two series of models, first, interacting the Back-to-Basic indicator with the four race, ethnicity and language groups and, second, interacting the Back-to-Basic indicator with the four service line groups. For each, we conducted the associated joint-F-test to test whether the effect was heterogenous across the groups.

To better understand the aspect of patient experience associated with Back-to-Basics, we fit OLS models similar to the main models, for all items comprising the Child HCAHPS measures that were significantly associated with Back-to-Basics implementation. Because patient length of stay can influence the depth and frequency of staff interactions and exposure to hospital processes, such as discharge planning and medication education, we conducted sensitivity analyses adding patient length of stay (calculated using admission date and discharge date) to the models as a covariate. All analyses were conducted using

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R version 4.4.2. Study protocols were approved by RAND's Human Subjects Protection Committee.

Results

Patient and Family Characteristics.

Parents and guardians reported similar mean child's age (8 years of age), child's general health (overall 80% in excellent or very good health), and relationship to the patient (overall 84% mothers). The distribution of the patient's race/ethnic/language group was different before and after Back-to-Basics, with 10% fewer responding as non-Hispanic white, 7% more responding as non-Hispanic black, and 4% more responding as Spanish-preferring Hispanic after Back-to Basics was implemented (Chi-sq test p-value <0.001). The distribution of parent/guardian's age and education level was also different before and after Back-to-Basics, with a shift towards younger and less educated respondents after Back-to-Basics was implemented (Chi-square test p-values <0.001, 0.004 respectively). Table 2 provides patient and respondent characteristics before and after Back-to-Basics implementation.

Improvements in Child HCAHPS measures.

The overall case-mix adjusted top-box scores were significantly higher after implementation of Back-to-Basics for five of the eight targeted Child HCAHPS measures--three measures related to communication, one related to mistake prevention and one about helping the child feel comfortable. The unadjusted results were similar to the adjusted results, except for one measure, *Keeping informed about child's care*, which in the unadjusted models had a 4pp increase from 82 to 86 that was statistically significant (p-value=0.032) and in the adjusted results had a 3pp increase from 82 to 85 that was not significant (p-value=0.193) (See Supplemental Table 1). The sensitivity analyses, which added patient length of stay as a covariate to the main models, demonstrated similar results to the adjusted main model results (Data not shown). Table 3 shows the overall mean case-mix adjusted top-box score for each of the Child HCAHPS measures before and after Back-to-Basics implementation.

To further understand what is driving the associations with the Back-to-Basics intervention and the significantly improve Child HCAHPS measures, we also assessed each item within those five improved Child HCAHPS measures. Table 4 shows the overall mean case-mix adjusted top-box scores for the items comprising each of the significant Child HCAHPS measures (identified in Table 3).

Communication. After implementation of Back-to-Basics, three communication measures improved: *Communication between you and your child's nurses* (mean 0.91 during the Back-to-Basics period, a 3 percentage point (pp) increase from before Back-to-Basics, p-value=0.04), *How well nurses communicate with your child* (mean 0.87 during the Back-to-Basics period, a 7 pp increase from before Back-to-Basics, p-value=0.001), and *How well doctors communicate with your child* (mean 0.86 during the Back-to-Basics period, a 6pp increase from before Back-to-Basics, p-value=0.007).

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For the measure *Communication between you and your child's nurses*, only one item, "During this hospital stay, how often did your child's nurses explain things to you in a way that was easy to understand?", was significantly associated with Back-to-Basics implementation, with 91% responding to the top-box category "Always" after Back-to-Basics implementation (i.e., mean 0.91) and a 5pp increase from before Back-to-Basics (p-value=0.008).

For the measure *How well nurses communicate with your child*, all three items were significantly associated with higher mean top-box scores after implementation of Back-to-Basics. The item "During this hospital stay, how often did your child's nurses encourage your child to ask questions?" was significantly associated with Back-to-Basics implementation, with a 10pp increase to 79% responding "Always" (p-value=0.002). The item "During this hospital stay, how often did your child's nurses listen carefully to your child?" showed a significant 5pp increase in the top-box category from 87% to 91% (p-value=0.041) and the item "During this hospital stay, how often did your child's nurses explain things in a way that was easy for your child to understand?" exhibited a significant 6pp increase, from 84% to 91% in the top-box category following Back-to-Basics implementation (p-value=0.011).

For the measure *How well doctors communicate with your child*, two of three items were associated with higher mean top-box scores after Back-to-Basics implementation. The item "During this hospital stay, how often did your child's doctors explain things in a way that was easy for your child to understand?" was significantly associated with Back-to-Basics implementation, with a 6pp increase to 89% responding "Always" (p-value=0.016) and item "During this hospital stay, how often did your child's doctors encourage your child to ask questions?" showed a significant 8pp increase in the mean top-box score from 73% to 81% (p-value=0.021).

Mistake Prevention. After implementation of Back-to-Basics, *Preventing mistakes and helping you report concerns* improved (mean 0.65 during the Back-to-Basics period, a 4pp increase from before Back-to-Basics, p-value=0.03). For the measure *Preventing mistakes and helping you report concerns*, only one item was significantly associated with an increase in mean top-box scores after Back-to-Basics implementation. The item "Before giving your child any medicine, how often did providers or other hospital staff check your child's wristband or confirm his or her identity in some other way?" was significantly associated with Back-to-Basics implementation; a 3pp increase to a 96% mean top-box score (p-value=0.019).

Helping Child Feel Comfortable. After implementation of Back-to-Basics, *Helping your child feel comfortable* improved (mean 0.77 during the Back-to-Basics period, a 4pp increase from before Back-to-Basics, p-value=0.03). For the measure *Helping your child feel comfortable*, only one item was significantly associated with an increase in mean top-box scores after Back-to-Basics implementation. The item "During this hospital stay, how often did providers talk with and act toward your child in a way that was right for your child's age?" was significantly associated with Back-to-Basics implementation; a 5pp increase to 89% mean top-box score (p-value=0.017).

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In the two series of models that tested whether the effect of Back-to-Basics differed by service line or by patient race, ethnicity and language groups, the findings indicated no significant differences across the four race, ethnicity and language groups or the four service line groups for all measures (Data not shown).

Nursing Turnover.

For context, in 2023 prior to the training period, the hospital's nursing turnover rate was 15.5% and after the training it dropped to 12.9% in 2024.

Discussion

Studies consistently show that effective communication between clinicians and patients is a critical determinant of overall patient experience.[24–33] Organizations engage in several ways to change providers behavior; many provide communication and safety training for clinicians.[20, 29, 30, 34–36] Our study is consistent with these studies in finding an immediate improvement directly following the Back-to-Basics training in mistake prevention, helping children feel comfortable, and communication (specifically, both nurse and doctor communication with the child, communication between the nurse and the parent/guardian of the child), as captured by five of the eight targeted Child HCAHPS measures. Notably, the changes in the scores for the remaining three targeted measures after Back-to-Basics were positive and in the magnitude of a 2pp or a 3pp increase; however, these increases were not statistically significant. Importantly, the training included a session for all employees that focused on strategies and resources for improving patient interactions, especially in challenging situations, and for those with non-English language preference; these focused on behaviors of listening, explaining, courtesy and respect, encouraging a child to ask questions, keeping parent/guardian informed about child's care including ER, test results, and procedures--which are behaviors captured in the targeted eight Child HCAHPS measures. The training included content and discussions for handling challenging situations, caring for those with non-English language preferences and the use of medical interpreters. With some additional emphasis on keeping parents, guardians and families informed about care in the ER and related to tests and procedures as well as for doctors and their listening, explaining, and treating parents and guardians with courtesy and respect, the training could improve the additional 3 targeted measures. The training did not significantly improve scores that were not emphasized in the training, which is to be expected. However, since communication is typically a driver of the global ratings, the increased scores in the 5 measures could have possibly impacted the global ratings. We do see improve global ratings in the magnitude of a 1pp and 3pp increase, but these increases were not at a magnitude that was statistically significant. For *Willingness to recommend*, because it was at 90 before the training there was not much room for improvement; measures that are close to the top of the performance range are also much harder to improve further. The *Overall rating of the hospital* had room for improvement and did increase 3pp, but that was not a large enough magnitude to be statistically significant even with a large sample.

Importantly, we also found no significant differences in the models that tested whether the effect of Back-to-Basics differed by service line or by patient racial, ethnic and language

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groups. This suggests that the effect of Back-to-Basics was uniform across patient groups and service lines. This points to the effectiveness of a hospital-wide training for all clinical and nonclinical staff, as this broad, inclusive approach to the training was able to impact all patient facing interactions. This allowed for there to be improved interactions across all service lines and patient groups, which was captured in the uniform improvements of scores across these groups of patients.

Limitations.

We studied one large, urban/suburban, freestanding children's hospital using Child HCAHPS data as the basis for evaluating a hospital-wide training that targeted areas in need of improvement and was provided to both clinical and nonclinical staff. The training focused on communication behaviors and safety protocols. As a single site study, our findings may not generalize to all settings of pediatric inpatient care, but the findings and intervention are instructive, given the limited research evaluating the effect of hospital-wide inpatient training focused on modifiable behaviors linked to patient experiences. We also were unable to evaluate all providers who completed the training, as patient experience surveys were not available for some providers before and after the training. Additionally, the overall response rate for the hospital is low, but not atypical for patient experience surveys. We minimized the risk of bias by employed case-mix adjustment. Further, our subgroup sample sizes may be limited in statistical power to be definitive of a uniform effect for the heterogeneity of treatment effect analysis. However, our findings support the current literature on the effectiveness of training but also extend the literature by linking the effectiveness of the training to specific Child HCAHPS outcomes.

In conclusion, a small investment of curricular time over three months devoted to nurse and doctor communication skills and safety procedures across all staffing roles improved Child HCAHPS communication-specific and safety scores. This training, which is focused on improving communication and safety skills and delivered to clinical and non-clinical staff, could be implemented in a variety of pediatric and inpatient settings to improve patient experience. Further examination of the intervention in different pediatric hospital sites, settings (i.e., hospital within a hospital), and contexts is warranted.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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References

1. Berwick DM. A user's manual for the IOM's 'Quality Chasm' report. Health Aff (Millwood). May-Jun 2002;21(3):80–90. doi:10.1377/hlthaff.21.3.80 [PubMed: 12026006]
2. Institute of Medicine. Crossing the quality chasm: A new health system for the 21st century. National Academy Press; 2001.
3. Elliott MN. Components of care vary in importance for overall patient-reported experience by type of hospitalization. Med Care. 2009:842–849. [PubMed: 19584764]
4. Giordano LA, Elliott MN, Goldstein E, Lehrman WG, Spencer PA. Development, implementation, and public reporting of the HCAHPS survey. Med Care Res Rev. Feb 2010;67(1):27–37. doi:10.1177/1077558709341065 [PubMed: 19638641]
5. Rothman AA, Park H, Hays RD, Edwards C, Dudley RA. Can additional patient experience items improve the reliability of and add new domains to the CAHPS hospital survey? Health Serv Res. Dec 2008;43(6):2201–22. doi:10.1111/j.1475-6773.2008.00867.x [PubMed: 18522668]
6. Weidmer BA, Brach C, Slaughter ME, Hays RD. Development of items to assess patients' health literacy experiences at hospitals for the Consumer Assessment of Healthcare Providers and Systems (CAHPS) Hospital Survey. Med Care. Sep 2012;50(9 Suppl 2):S12–21. doi:10.1097/MLR.0b013e31826524a0 [PubMed: 22895225]
7. Einsenberg J. Doctors' decisions and the cost to medical care. 1986.
8. Fineberg HV. Clinical evaluation: How does it influence medical practice? Bull Cancer. 1987;74(3):333–46. [PubMed: 3304481]
9. Associates D. Improving Care Team Communication & Patient Experience. 2015.
10. Davidson KW, Shaffer J, Ye S, et al. Interventions to improve hospital patient satisfaction with healthcare providers and systems: a systematic review. BMJ Qual Saf. Aug 3 2016;doi:10.1136/bmjqs-2015-004758
11. Seiler A, Knee A, Shaaban R, et al. Physician communication coaching effects on patient experience. PLoS One. 2017;12(7):e0180294. doi:10.1371/journal.pone.0180294 [PubMed: 28678872]
12. Wang W, Dudjak LA, Larue EM, Ren D, Scholle C, Wolf GA. The influence of goal setting and SmartRoom patient education videos on readmission rate, length of stay, and patient satisfaction in the orthopedic spine population. Comput Inform Nurs. Sep 2013;31(9):450–6. doi:10.1097/NXN.0b013e3182999df8 [PubMed: 23917646]
13. Boissy A, Windover AK, Bokar D, et al. Communication skills training for physicians improves patient satisfaction. J Gen Intern Med. Jul 2016;31(7):755–61. doi:10.1007/s11606-016-3597-2 [PubMed: 26921153]
14. Tackett S, Tad-y D, Rios R, Kisuule F, Wright S. Appraising the practice of etiquette-based medicine in the inpatient setting. J Gen Intern Med. Jul 2013;28(7):908–13. doi:10.1007/s11606-012-2328-6 [PubMed: 23423452]
15. Brosey LA, March KS. Effectiveness of structured hourly nurse rounding on patient satisfaction and clinical outcomes. J Nurs Care Qua. Apr-Jun 2015;30(2):153–9. doi:10.1097/NCQ.0000000000000086
16. Schroeder DL, Hoffman LA, Fioravanti M, Medley DP, Zullo TG, Tuite PK. Enhancing nurses' pain assessment to improve patient satisfaction. Orthop Nurs. Mar-Apr 2016;35(2):108–17; quiz 118-9. doi:10.1097/NOR.0000000000000226 [PubMed: 27028687]
17. Allenbaugh J, Corbelli J, Rack L, Rubio D, Spagnoletti C. A brief communication curriculum improves resident and nurse communication skills and patient satisfaction. J Gen Intern Med. Jul 2019;34(7):1167–1173. doi:10.1007/s11606-019-04951-6 [PubMed: 30997637]
18. Doyle C, Lennox L, Bell D. A systematic review of evidence on the links between patient experience and clinical safety and effectiveness. British Medical Journal Open. Jan 3 2013;3(1)doi:10.1136/bmjopen-2012-001570
19. Jun J, Stern K, Djukic M. Integrative review of the interventions for improving patients' experiences revealed in quality improvement projects. J Patient Exp. Dec 2020;7(6):882–892. doi:10.1177/2374373520925271 [PubMed: 33457515]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

20. Brown JB, Boles M, Mullooly JP, Levinson W. Effect of clinician communication skills training on patient satisfaction. A randomized, controlled trial. Ann Intern Med. Dec 7 1999;131(11):822–9. doi:10.7326/0003-4819-131-11-199912070-00004 [PubMed: 10610626]
21. OLeary KJ, Darling TA, Rauworth J, Williams MV. Impact of hospitalist communication-skills training on patient-satisfaction scores. J Hosp Med. Jun 2013;8(6):315–20. doi:10.1002/jhm.2041 [PubMed: 23554016]
22. NSI Nursing Solutions, Inc. 2025 NSI National Health Care Retention & RN Staffing Report. 2025. https://www.nsinursingsolutions.com/Documents/Library/NSI_National_Health_Care_Retention_Report.pdf
23. Quigley DD, Elliott MN, Slaughter ME, Hays RD. Using a quality improvement priority metric to promote uniformly high-quality care: An illustrative example using Child HCAHPS Survey data. J Healthc Manag. 2025, under review;
24. Carter WB, Inui TS, Kukull WA, Haigh VH. Outcome-based doctor-patient interaction analysis: II. Identifying effective provider and patient behavior. Med Care. Jun 1982;20(6):550–66. doi:10.1097/00005650-198206000-00002 [PubMed: 7109739]
25. Davis DA, Thomson MA, Oxman AD, Haynes RB. Changing physician performance. A systematic review of the effect of continuing medical education strategies. JAMA. Sep 6 1995;274(9):700–5. doi:10.1001/jama.274.9.700 [PubMed: 7650822]
26. Frederickson L. Exploring information-exchange in consultation: the patients' view of performance and outcomes. Patient Educ Couns. 1995;25:237–46. [PubMed: 7630827]
27. Hall JA, Irish JT, Roter DL, Ehrlich CM, Miller LH. Satisfaction, gender, and communication in medical visits. Med Care. Dec 1994;32(12):1216–31. doi:10.1097/00005650-199412000-00005 [PubMed: 7967860]
28. Hall JA, Roter DL, Katz NR. Meta-analysis of correlates of provider behavior in medical encounters. Med Care. Jul 1988;26(7):657–75. doi:10.1097/00005650-198807000-00002 [PubMed: 3292851]
29. Joos SK, Hickam DH, Gordon GH, Baker LH. Effects of a physician communication intervention on patient care outcomes. J Gen Intern Med. Mar 1996;11(3):147–55. doi:10.1007/BF02600266 [PubMed: 8667091]
30. Maiman LA, Becker MH, Liptak GS, Nazarian LF, Rounds KA. Improving pediatricians' compliance-enhancing practices. A randomized trial. Am J Dis Child. Jul 1988;142(7):773–9. doi:10.1001/archpedi.1988.02150070087033 [PubMed: 3381783]
31. Mishler EG, Clark JA, Ingelfinger J, Simon MP. The language of attentive patient care: A comparison of two medical interviews. J Gen Intern Med. Jul-Aug 1989;4(4):325–35. doi:10.1007/BF02597407 [PubMed: 2668459]
32. Roter DL, Stewart M, Putnam SM, Lipkin M Jr., Stiles W, Inui TS. Communication patterns of primary care physicians. JAMA. Jan 22-29 1997;277(4):350–6. [PubMed: 9002500]
33. Rowland-Morin PA, Carroll JG. Verbal communication skills and patient satisfaction. A study of doctor-patient interviews. Eval Health Prof. Jun 1990;13(2):168–85. doi:10.1177/016327879001300202 [PubMed: 10106792]
34. Fustino NJ, Moore P, Viers S, Cheyne K. Improving patient experience of care providers in a multispecialty ambulatory pediatrics practice. Clin Pediatr (Phila). Jan 2019;58(1):50–59. doi:10.1177/0009922818806309 [PubMed: 30296841]
35. Lewis CC, Pantell RH, Sharp L. Increasing patient knowledge, satisfaction, and involvement: randomized trial of a communication intervention. Pediatrics. Aug 1991;88(2):351–8. [PubMed: 1861939]
36. Weech-Maldonado R, Morales LS, Spritzer K, Elliott M, Hays RD. Racial and ethnic differences in parents' assessments of pediatric care in Medicaid managed care. Health Serv Res. Jul 2001;36(3):575–94. [PubMed: 11482590]

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Table 1.

Back-to-Basic Content, Delivery Style, Learning Objectives, and Included Roles, by Module

Module	Content	Delivery Style	Learning Objectives	Included Roles
Patient Experience	Explored differences between patient satisfaction, perception, and overall experience, emphasizing their impact on the hospital's mission. Highlighted strategies and resources for improving patient interactions, especially in challenging situations, and for those with non-English language preference; these focused on behaviors of listening, explaining, courtesy and respect, encouraging a child to ask questions, keeping parent/guardian informed about child's care including ER, test results, and procedures and use of medical interpreters.	Mini lecture, video, Q&A	- Define and differentiate between patient satisfaction, perception, and experience. - Explain the importance of patient experience in relation to the hospital's mission. - Identify components that contribute to a positive patient experience. - Discuss strategies for handling challenging patient interactions emphasizing listening, explaining, courtesy and respect, encouraging a child to ask questions, keeping parent/guardian informed about child's care including ER, test results, and procedures. - Discuss strategies for visits with non-English language preference and use of medical interpreters Locate resources that support staff in enhancing patient experience.	Clinical and Non-Clinical
Environment of Care	Focused on maintaining a safe, clean, and therapeutic environment that supports patient healing and staff wellbeing through adherence to protocols, regulatory compliance and teamwork.	Discussion, Q&A	- Describe key elements that contribute to a safe and healing environment. - Discuss how staff actions impact the environment of care. - Identify resources and protocols related to maintaining a safe environment.	Clinical and Non-Clinical
Journey to Excellence/ Magnet®	Reviews the Magnet® Recognition Program and hospital's professional practice model, showcasing their influence on patient, family, and staff experiences as part of the journey to healthcare excellence.	Discussion, Q&A, gamification	- Explain purpose and goals of Magnet® Recognition Program. - Describe hospital's professional practice model and its role in improving patient, family, and staff experiences. - Recognize the impact of Magnet® designation on patient care and outcomes. - Interpret the significance of the hospital's Magnet® journey and site visit expectations.	Clinical and Non-Clinical
Infection Control	Reviews essential infection prevention practices, including hand hygiene and surgical preparation, aimed at achieving zero patient harm related to infections.	Discussion, demonstration, gamification Q&A	- Describe practices that support safe patient care to achieve 'Zero Harm.' - Demonstrate proper hand hygiene according to the 5 Moments model. - Explain infection prevention steps during surgical patient preparation.	Clinical only
Central Line Basics	Covers evidence-based techniques for central line maintenance to reduce infection risks, including the proper "scrub the hub" method.	Discussion, demonstration, gamification Q&A	- Identify evidence-based practices for central line care. - Demonstrate the "scrub the hub" technique correctly. - Explain the rationale for each step in central line maintenance.	Clinical only
Labs	Outlined procedures for specimen collection, correct use of lab devices, and safe transport, and integrating these practices with the hospital's electronic medical records system.	Discussion and Demonstration, Q&A	- Locate and apply the minimum specimen volume list in the electronic health record. - Identify appropriate lab collection devices and tubes. - Demonstrate safe specimen transport practices using the tube system.	Clinical only
Pressure Injuries	Detailed early detection, risk factors, and treatment strategies for healthcare-acquired pressure injuries through thorough skin and wound assessments.	Discussion and Demonstration, Q&A, gamification	- Identify risk factors for healthcare-acquired pressure injuries. - Demonstrate best practices in skin and wound assessment. - Describe early detection and intervention strategies.	Clinical only
National Database of Nursing Quality Indicators and	Examined the collection and interpretation of nursing-sensitive quality indicators and patient experience data to inform quality	Presentation, Discussion and Gamification	- Interpret nursing-sensitive clinical indicator data. - Analyze patient experience data. - Explain the relevance of data trends to	Clinical and Non-Clinical

Module	Content	Delivery Style	Learning Objectives	Included Roles
nursing sensitive Patient Experience data	improvement efforts across all roles.		quality improvement initiatives, regardless of role.	

NOTE: "All" employees includes employees such as environmental services (EVS), security, kitchen staff etc.

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Table 2.

Patient and Child HCAHPS Survey Respondent Characteristics Before and After Back-to-Basics Training Implementation

Characteristic	Before Back-to-Basics (5/1/21 to 9/19/23) N=544	After Back-to-Basics (9/20/23 to 10/15/24) N=653	Overall N=1,197	P-value
Child's Age, Mean (SD)	8 (6)	8 (6)	8 (6)	0.7
Race, Ethnicity and Language, N (%)				**<0.001**
Non-Hispanic White	183 (34%)	157 (24%)	340 (28%)	
English-preferring Hispanic	178 (33%)	207 (32%)	385 (32%)	
Spanish-preferring Hispanic	83 (15%)	124 (19%)	207 (17%)	
Non-Hispanic Black	100 (18%)	165 (25%)	265 (22%)	
General Health of Child, N (%)				0.2
Excellent	256 (48%)	325 (50%)	581 (49%)	
Very good	165 (31%)	196 (30%)	361 (31%)	
Good	78 (15%)	100 (15%)	178 (15%)	
Fair/Poor	33 (6.2%)	30 (4.6%)	63 (5.3%)	
Unknown	12	2	14	
Respondent's Age, N (%)				**<0.001**
24 years old or younger	20 (3.8%)	62 (9.6%)	82 (7.0%)	
25-34	94 (18%)	140 (22%)	234 (20%)	
35-44	227 (43%)	269 (42%)	496 (42%)	
45-54	152 (29%)	142 (22%)	294 (25%)	
55 years of age or older	38 (7.2%)	34 (5.3%)	72 (6.1%)	
Unknown	13	6	19	
Respondent's Education, N (%)				**0.004**
Less than high school	23 (4.4%)	43 (6.7%)	66 (5.7%)	
High school graduate	44 (8.4%)	89 (14%)	133 (11%)	
Some college	128 (24%)	169 (26%)	297 (26%)	
4-yr coll. graduate	133 (25%)	139 (22%)	272 (23%)	
More than 4 years college	195 (37%)	200 (31%)	395 (34%)	
Unknown	21	13	34	
Respondent's Relationship to Patient, N (%)				0.9
Mother	449 (84%)	545 (84%)	994 (84%)	
Other relationship	86 (16%)	102 (16%)	188 (16%)	
Unknown	9	6	15	
Pediatric Service Line (from administrative data)				0.4
Pediatrics General	262 (48%)	313 (48%)	575 (48%)	
Intensive/Critical Care	77 (14%)	100 (15%)	177 (15%)	
Surgery	105 (19%)	105 (16%)	210 (18%)	
Specialties	100 (18%)	135 (21%)	235 (20%)	

NOTES: HCAHPS stands for Hospital Consumer Assessment of Healthcare Providers and Systems. SD indicates standard deviation. Race, ethnicity and language indicates self-reported preferred language of the respondent and the child's race and ethnicity and is categorized into four groups: English-preferring Hispanic, Spanish-preferring Hispanic, non-Hispanic Black, and non-Hispanic White patients. Chi-sq tests were used to

compare before and after Back-to-Basics patient/respondent characteristics for all variables except Child's Age which used a Wilcox test. P-values < 0.05 are in bold.

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Table 3.

Mean Case-Mix Adjusted Top-box Scores for Child HCAHPS Measures Targeted (or Not) for Improvement, Before and After Implementation of Back-to-Basics Training

Child HCAHPS Measures		Before Back-to-Basics	After Back-to-Basics
Eight Measures Targeted for Improvement	**N**	**Mean Adjusted Top-box Score (95% CI)**	**Mean Adjusted Top-box Score (95% CI)**
Composites			
Communication between you and your child's nurses	1,196	0.88 (0.86, 0.90)	**0.91 (0.89, 0.93)** *
Communication between you and your child's doctors	1,193	0.89 (0.87, 0.91)	0.91 (0.89, 0.93)
Keeping you informed about your child's care#	1,191	0.82 (0.80, 0.85)	0.85 (0.82, 0.87)
How well nurses communicate with your child#	683	0.80 (0.76, 0.83)	**0.87 (0.84, 0.90)** **
How well doctors communicate with your child#	684	0.80 (0.77, 0.83)	**0.86 (0.83, 0.89)** **
Preventing mistakes and helping you report concerns#	1,186	0.61 (0.58, 0.63)	**0.65 (0.63, 0.67)** *
Helping your child feel comfortable	1,192	0.73 (0.70, 0.75)	**0.77 (0.74, 0.79)** *
Single-Items			
Keeping you informed about your child's care in the ER	673	0.90 (0.87, 0.93)	0.92 (0.90, 0.95)
Eight Measures & Two Global Ratings Not Targeted for Improvement	**N**	**Mean Adjusted Top-box Score (95% CI)**	**Mean Adjusted Top-box Score (95% CI)**
Composites			
Communication about your child's medicines#	1,153	0.79 (0.77, 0.82)	0.82 (0.79, 0.84)
Preparing you and your child to leave the hospital	1,189	0.85 (0.83, 0.88)	0.86 (0.84, 0.88)
Involving teens in their care	345	0.79 (0.74, 0.85)	0.74 (0.69, 0.79)
Single-Items			
Privacy when talking with doctors, nurses, other providers	1,190	0.90 (0.88, 0.93)	0.91 (0.89, 0.93)
Responsiveness to call button	831	0.78 (0.74, 0.82)	0.81 (0.78, 0.85)
Paying attention to your child's pain	773	0.83 (0.79, 0.87)	0.83 (0.79, 0.86)
Cleanliness of hospital room	1,186	0.81 (0.78, 0.85)	0.83 (0.80, 0.86)
Quietness of hospital room	1,181	0.74 (0.70, 0.78)	0.75 (0.72, 0.79)
Global Ratings			
Overall rating of hospital	1,187	0.85 (0.83, 0.88)	0.88 (0.85, 0.90)
Willingness to recommend the hospital	1,187	0.90 (0.88, 0.92)	0.91 (0.89, 0.94)

Notes: HCAHPS stands for Hospital Consumer Assessment of Healthcare Providers and Systems. CI indicates confidence intervals.

indicates the five Child HCAHPS Leapfrog measures (See https://ratings.leapfroggroup.org/sites/default/files/2024-03/2024%20Pediatric%20Care%20Fact%20Sheet.pdf). Time periods are before Back-to-Basics (5/1/2021 to 9/19/2023) and after Back-to-Basics (9/20/2023 to 10/15/2024). Models case-mix adjust for service line group (Pediatric General, Intensive/Critical Care, Surgery, Specialty),race/ethnicity/language (Non-Hispanic White, Non-Hispanic Black, Hispanic English preferring, Hispanic Spanish preferring), age of child, child's general health (Excellent, Very Good, Good, Fair/Poor), respondent's age (<25 yrs old, 25 to 34, 35 to 44, 45 to 54, 55 to 64, 65 or older), education (Less than High School, High School graduate, Some college, 4-year college graduate) and relationship to child (Mother, Other than Mother). Stars represent p-values from t tests comparing means before and after implementation of Back-to-Basics; **Bold** indicates statistical significance at

* $p < 0.05$,

** $p < 0.01$ based on an OLS regression model that included an independent 0/1 indicator for Back-to-Basics.

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Table 4.

Mean Case-Mix Adjusted Top-box Scores for Items within Significant Child HCAHPS Composites, Before and After Implementation of Back-to-Basics Training

Child HCAHPS Composite Measures & Items	N	Before Back-to-Basics Mean (95% CI)	After Back-to-Basics Mean (95% CI)
Communication between you and your child's nurses [+]			
During this hospital stay, how often did child's nurses listen carefully to you?	1,193	0.86 (0.83, 0.88)	0.88 (0.86, 0.91)
During this hospital stay, how often did child's nurses explain things to you in a way that was easy to understand?	1,194	0.87 (0.84, 0.89)	**0.91 (0.89, 0.94)** **
During this hospital stay, how often did child's nurses treat you with courtesy and respect?	1,194	0.92 (0.89, 0.94)	0.93 (0.91, 0.95)
How well nurses communicate with your child [+#]			
During this hospital stay, how often did child's nurses encourage your child to ask questions?	680	0.68 (0.63, 0.73)	**0.79 (0.74, 0.83)** **
During this hospital stay, how often did child's nurses listen carefully to your child?	682	0.87 (0.83, 0.90)	**0.91 (0.88, 0.95)** *
During this hospital stay, how often did child's nurses explain things in a way that was easy for your child to understand?	682	0.84 (0.81, 0.88)	**0.91 (0.87, 0.94)** *
How well doctors communicate with your child [+#]			
During this hospital stay, how often did child's doctors listen carefully to your child?	682	0.84 (0.81, 0.88)	0.89 (0.86, 0.93)
During this hospital stay, how often did child's doctors explain things in a way that was easy for your child to understand?	681	0.82 (0.79, 0.86)	**0.89 (0.85, 0.92)** *
During this hospital stay, how often did child's doctors encourage your child to ask questions?	675	0.73 (0.69, 0.78)	**0.81 (0.76, 0.85)** *
Preventing mistakes and helping you report concerns [+#]			
Before giving your child any medicine, how often did providers or other hospital staff check child's wristband or confirm his or her identity in some other way?	1,103	0.93 (0.91, 0.95)	**0.96 (0.95, 0.98)** *
During this hospital stay, did providers or other hospital staff tell you how to report if you had any concerns about mistakes in your child's health care?	1,172	0.32 (0.28, 0.36)	0.37 (0.33, 0.40)
Helping your child feel comfortable [+]			
Things that a family might know best about a child include how the child usually acts, what makes the child comfortable, and how to calm the child's fears. During this hospital stay, did providers ask you about these types of things?	1,183	0.61 (0.57, 0.65)	0.64 (0.61, 0.68)
During this hospital stay, how often did providers talk with and act toward your child in a way that was right for child's age?	1,190	0.84 (0.81, 0.87)	**0.89 (0.86, 0.92)** *
Hospitals can have things like toys, books, mobiles, and games for children from newborns to teenagers. During this hospital stay, did the hospital have things available for your child that were right for your child's age?	1,162	0.73 (0.69, 0.77)	0.76 (0.73, 0.80)

Notes: HCAHPS stands for Hospital Consumer Assessment of Healthcare Providers and Systems. CI indicates confidence intervals.

[+] indicates measures targeted for improvement.

[#] indicates the five Child HCAHPS Leapfrog measures (See https://ratings.leapfroggroup.org/sites/default/files/2024-03/2024%20Pediatric%20Care%20Fact%20Sheet.pdf). Time periods are before Back-to-Basics (5/1/2021 to 9/19/2023) and Back-to-Basics (9/20/2023 to 10/15/2024). Models case-mix adjust for service line group (Pediatric General, Intensive/Critical Care, Surgery, Specialty), race/ethnicity/language (Non-Hispanic White, Non-Hispanic Black, Hispanic English preferring, Hispanic Spanish preferring), age of child, child's general health (Excellent, Very Good, Good, Fair/Poor), respondent's age (<25 yrs old, 25 to 34, 35 to 44, 45 to 54, 55 to 64, 65 or older),

education (Less than High School, High School graduate, Some college, 4-year college graduate) and relationship to child (Mother, Other than Mother). Stars represent p-values from t tests comparing means before and after implementation of Back-to-Basics; **Bold** indicates statistical significance at

*
$p < 0.05$,

**
$p < 0.01$,

***
$p < 0.001$ based on an OLS regression model that included an independent 0/1 indicator for Back-to-Basics.

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